

Workshop 3

1. Consider a survey in which a random sample of vegetarians and non-vegetarians was taken, and their weight y_i , height x_i , and gender were recorded. A model for the resulting data might be

$$\mathbb{E}(y_i) = \alpha + \nu_k + \gamma_j + \beta x_i \text{ if } y_i \text{ is for diet } k \text{ and gender } j, \quad j \in \{1, 2\}, \quad k \in \{1, 2\},$$

where $k = 1$ for vegetarian, $k = 2$ for non-vegetarian, $j = 1$ for female, $j = 2$ for male, $y_i \sim N(\mu_i, \sigma^2)$, $i = 1, \dots, n$, and all Greek letters refer to model parameters.

- (a) Write out the model matrix (design matrix) and parameter vector for this model, ignoring identifiability problems. For the sake of concreteness assume a random sample of $n = 8$ subjects, the first four vegetarian, next four not, with the first two of each set of four being female and the next two male.
- (b) Re-write the model matrix and parameter vector so that the model is identifiable.

2. (a) Consider the linear model

$$y_{ij} = \alpha + \beta_i + \gamma_j + \delta z_{ij} + \epsilon_{ij}, \quad i = 1, \dots, 3, \quad j = 1, \dots, 4,$$

where α , β_i , γ_j , and δ are parameters, y_{ij} and z_{ij} are variables, and the ϵ_{ij} are iid $N(0, \sigma^2)$. Write the model in the matrix-vector form $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, without worrying about identifiability constraints.

- (b) Rewrite your model from part (a) with identifiability constraints.
- (c) In the following call to the R function `lm`, `a` and `b` are factor variables, while `v` and `w` are continuous variables.

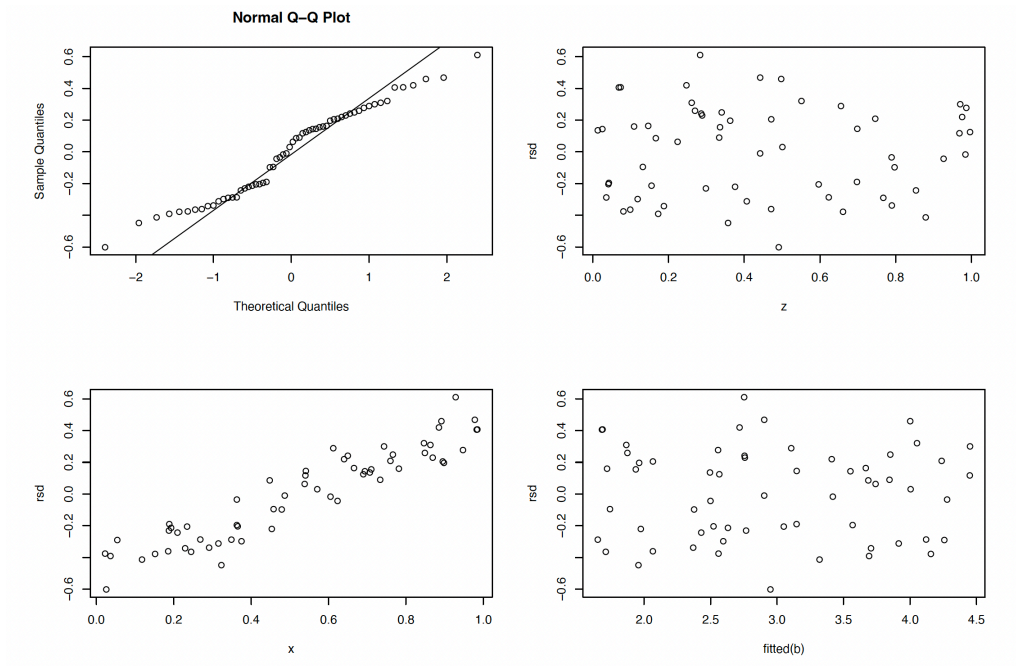
```
lm(y ~ a*b + v + w + I(v*w))
```

Write the mathematical form of the model being fitted, as an equation and distributional assumption, without worrying about identifiability constraints. You should not write the model in matrix-vector form this time.

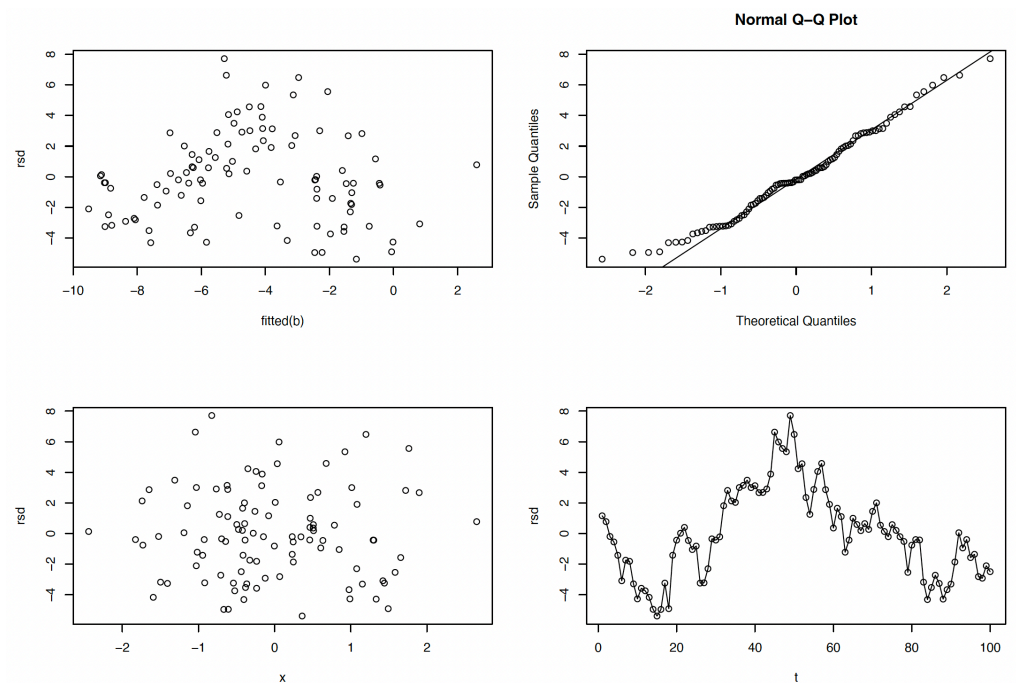
3. In each part of this question, some R code is used to fit a linear model to a dataset and generate the residual plots shown. In both cases, describe the main problem (if any) revealed by the plots and, if possible, suggest how it might be addressed.

(a)

```
names(dat)
"y" "x" "z" "fac"
b <- lm(y ~ fac + z, data = dat)
rsd <- residuals(b)
par(mfrow = c(2,2))
qqnorm(rsd); qqline(rsd)
with(dat, plot(z, rsd))
with(dat, plot(x, rsd))
plot(fitted(b), rsd)
```



```
(b) b <- lm(y ~ x + t)
     rsd <- residuals(b)
     par(mfrow = c(2,2))
     plot(fitted(b), rsd)
     qqnorm(rsd); qqline(rsd)
     plot(x, rsd)
     plot(t, rsd); lines(t, rsd)
```



4. This question is about model checking, transformation and using factor variables in R. You will use the `InsectSprays` data supplied with R. The data frame contains `counts` of pest insects on plots of an agricultural experiment which had each been treated with one of 6 insecticide formulations (the factor variable `spray` indicates which). Interest is in whether formulation makes a difference to the final count, and if so, which formulation is best.

- (a) Produce a plot of counts against spray type. How would you interpret the plot?
- (b) Fit a linear model to the data, with `count` as the response and `spray` as the predictor and check the residuals against the fitted values (or the square root of the absolute value of the standardised residuals against the fitted values). Is the constant variance assumption of the linear model met?
- (c) Is the normality assumption tenable for this linear model?
- (d) Count data are often more naturally modelled as following a Poisson distribution, rather than a normal. Using the Normal as an approximation to Poisson (or a transformation of Poisson) is usually OK, provided that we can somehow fix the problem that Poisson data do not have constant variance. Transformation can often do this (at least, approximately). It can then be shown that we should be able to approximately meet the constant variance assumption by using square root of `count` as the response, rather than `count`. Please try this. Does it improve matters?
- (e) Now look at a summary of your model. What is the interpretation of the `(Intercept)` parameter here? What about the other parameters?
- (f) Sometimes it is necessary to change the ‘reference’ level of a factor variable. For example, suppose that you wanted to compare all the formulations to formulation C. As shown in the notes, you can do this by using the `relevel` function:

```
InsectSprays$spray <- relevel(InsectSprays$spray, "C")
```

and refitting the model. Please try it and look at the summary output again: how has it changed? What would you conclude about the data now?

- (g) Finally, try refitting the model to the untransformed counts, with C as the reference level. Look at how the model summary changes relative to the transformed counts. Notice how your conclusions are completely different: why is this, and which answers would you trust?

5. The *High School and Beyond* dataset is available in `hsb` in the R package `faraway` (which you must install on your computer). Data was collected as a subset of the *High School and Beyond* study conducted by the National Education Longitudinal Studies program of the National Center for Education Statistics. `hsb` is a data frame with 200 observations and 11 variables. To know more about the variables, please type `help(hsb)`. One purpose of the study was to determine which factors influence students’ choice of high school program type—academic, vocational, or general. Since we have not yet learned the necessary tools to fit a model for this investigation, we will follow Faraway (2014, Chapter 16) and shift our focus slightly.

- (a) Model the math score as a function of the following five factors: gender, race, socioeconomic class, school type, and choice of high school program. Include all second-order interactions but no higher-order interactions. Implement the model in R and check its underlying assumptions. **Hint:** A model including all second order interactions between predictor variables, say, `u`, `v`, `w`, and `z`, can simply be fitted as

```
lm(y ~ (u + v + w + z)^2)
```

- (b) Perform backward model selection (you can use, for instance, the `step` function) and write down your final model using the identifiability constraints used in R by default. Do not forget to also check whether your final model does not violate any of the underlying assumptions of the linear model.
 - (c) Briefly report your findings based on the model obtained in (b).
6. The dataset `teengamb` in the package `faraway` concerns a study of teenage gambling in Britain. To know more about the variables in this dataset, please type `help(teengamb)`.
- (a) Fit a regression model with the expenditure on gambling as the response and the sex, socioeconomic status, income, and verbal score as predictors. Fit the same model, but now with the response variable being the square root of the expenditure on gambling. Why should the second model be preferred over the first model?
For the remaining questions, please use the square root of the expenditure as the response variable.
 - (b) With all other predictor variables held constant, what would be the estimated difference in the mean (square root) expenditure on gambling for a male compared to a female. Provide also a 95% confidence interval for this difference.
 - (c) Predict the amount that a male with socioeconomic status, income, and verbal score at the maximum values in this dataset would gamble, along with a 95% prediction interval. Take care to give your answer in the original units of the response.
 - (d) Fit a model with income just as a predictor, and use an F-test to compare it to the full model fitted in (a). State precisely the two models being compared. What does this tell you about the significance of the other three predictor variables?
 - (e) A straightforward model selection strategy in this context is to consider all possible subsets of predictors. Assuming sex must be included in all models and excluding interactions for simplicity, fit all possible combinations of the remaining predictors. How stable is the estimated effect of sex across these models?
 - (f) Which of the models fitted in (e) would you select as your final one, and on what basis? What conclusions would you draw from this selected model?